

AMG173 SERIES INDUSTRIAL GIGABIT ETHERNET HIGH SPEED VDSL2 EXTENDERS



Industrial Ethernet Solutions

AMG's industrial high-speed VDSL2 Ethernet Extenders provide transmission of high-speed Ethernet data over legacy cabling infrastructure including UTP, Coax, Alarm, Bell or Telephone cables.



					
Gigabit x1	VDSL2 x1	Temp -20~+65°C	Mounting DIN/Surface	Protection IP30	NDAA/TAA Compliant

[AMG173-1G-1V]

/ OVERVIEW

The AMG173 series are industrial high speed Gigabit Ethernet VDSL2 extenders that support an aggregated bandwidth up to 300Mbps (Downstream 150Mbps, Upstream 150Mbps).

The units feature a Gigabit Ethernet port with an RJ45 connector and a high speed VDSL2 port with an RJ45 connector (coax and screw terminal adapters are also included in the package) in a rugged metal housing to provide reliable operation in harsh environments.

The units are completely transparent to protocols, codes, and applications ensuring compatibility with any IP camera and its management software or any other IP device and operate in a simple unmanaged mode.

It is a perfect solution for sending video links from remote camera installations which are beyond the 100m (328ft) distance limit of Ethernet standards.

Support for Symmetric and Asymmetric profiles is included providing the option to maintain equal speeds in both directions or to have higher speeds in one direction for installations with IP cameras or other devices with high traffic flow in one direction.

/ FEATURES

- Compact size – ideal for confined spaces, including camera poles and roadside cabinets
- Gigabit Ethernet port supports high speed VDSL2
- High speed Ethernet extension over UTP, Coax, CAT 5e/6/7, Alarm or Telephone cables (1 Pair)
- Supports VDSL2 profiles 35b / 30a / 17a
- Distances up to 2000m (6560ft) over coax cable or CAT5e/6 UTP cable
- IEEE 802.1Q VLAN tag transparent
- Supports 8 selectable profiles (G.INP/Interleaved High & Low, Target SNR 8/10 dB, Symmetric/ Asymmetric modes)
- Compatible with J-Y(ST)Y 4x2x0.8 or J-Y(ST)Y 6x2x0.6 etc twisted pair cable
- Integrated surge protection
- AMG Lifetime Support Warranty

Specifications.

Standards.

IEEE802.3i	10Base-T
IEEE802.3u	100Base-TX
IEEE802.3ab	1000Base-T
ITU-T G993.1, G993.2, G993.5, G997.1, G998, G.INP	VDSL DMT Encoding

Interface.

LED Indicators	Power Remote (RT) VDSL2 Link/Activity Ethernet Link/Activity Ethernet Speed
Ethernet Ports	1x 10/100/1000T(X) RJ45 with Auto MDI/MDI-X and 1.5 kV Isolation Protection
VDSL2 Ports	1x RJ45 with Surge Protection (2 pins used - 4 & 5)
Power	1x 2 pin screw terminal block

Dip Switch.

Switch Type	4 Position
Functions	Central (Master) or Remote (RT) 8 VDSL2 Profiles: (G.INP, Interleave, Target SNR 8/10dB, Symmetric / Asymmetric Modes)

Power.

Power Inputs	1
Operating Voltage	9-30V _{DC}
Power Consumption	4.5W Max
Protection	Reverse Polarity Overload Current

Packaging.

Shipping Weight	0.43kg / 0.95lb
Dimensions	(W x D x H) 220 x 170 x 40 mm 8.66 x 6.69 x 1.57 in

Mechanical.

Housing	Metal
Dimensions:	(W x D x H) 95 x 73 x 23 mm 3.74 x 2.87 x 0.91 in
IP Rating	IP30
Installation	Wall Mount or DIN-Rail
Weight	0.22kg / 0.49lb

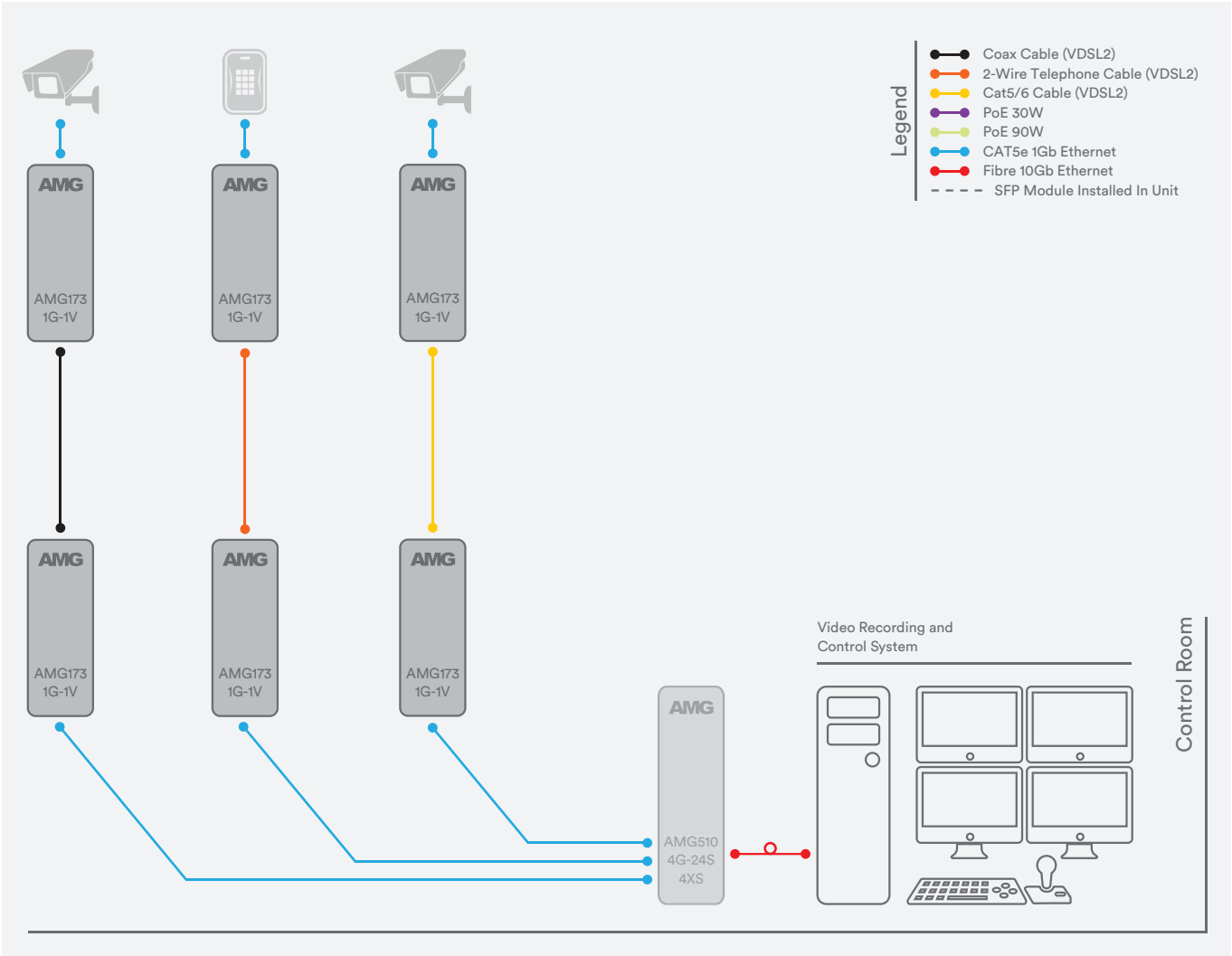
Environmental.

Operating Temp.	-20 to +65°C / -4 to +149°F
Storage Temp.	-40 to +70°C / -40 to +158°F
Humidity	0% to 95% (non-condensing)
MTBF	799,011 hours
MTBF Standard	MIL-HDBK-217F GB 25°C
Heat Dissipation	15 BTU/h
Cooling	Passive Cooling
Noise Level	0 dBA

Regulatory.

Safety	IEC/EN 62368-1
EMI	EN 55032 Class A CISPR 32 EN 300 386 FCC Part 15B Class A EN 61000-3-2 EN 61000-3-3 EN 55024 CISPR 24 EN 61000-4-2 (ESD) EN 61000-4-3 (RS) EN 61000-4-4 (EFT) EN 61000-4-5 (Surge) EN 61000-4-6 (CS) EN 61000-4-8 (PFMF) EN 61000-4-11 (Dip)
EMS	IEC 60068-2-27 IEC 60068-2-32 IEC 60068-2-6
Shock	Reach
Free Fall	RoHS
Vibration	WEEE
Environmental	NDAA & TAA Compliant
Supply Chain	

Application Diagram.



Data Rate Performance.

UTP Cable - 26AWG (0.45mm)			
Profile Setting: Symmetric , SNR Low, G.INP			
Distance (m)	Distance (ft)	Central to RT Speed (Mbps)	RT to Central Speed (Mbps)
152	500	162	146
305	1000	112	116
610	2000	61	33
914	3000	30	28
1219	4000	26	15
1524	5000	16	10
1828	6000	8	9

UTP Cable - 26AWG (0.45mm)			
Profile Setting: Asymmetric , SNR Low, Interleave			
Distance (m)	Distance (ft)	Central to RT Speed (Mbps)	RT to Central Speed (Mbps)
152	500	284	64
305	1000	171	59
610	2000	67	31
914	3000	35	10
1219	4000	20	5
1524	5000	28	1
1828	6000	18	1

The performance data shown in these tables is for reference only, the actual data rate may vary depending on the quality of the copper cable as well as environmental factors.

When using Asymmetric mode the maximum speed will be achieved in the direction from Central (Master) device to RT device as shown above.

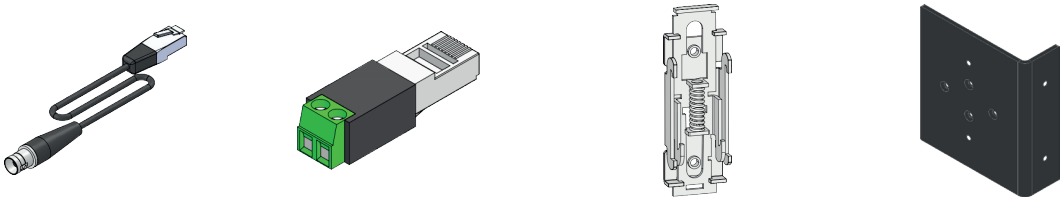
Part Numbers.

Industrial Gigabit Ethernet VDSL2 Extenders

AMG173-1G-1V	1 x 10/100/1000BaseT(x) RJ45, 1 x VDSL2 RJ45 For UTP, Coax or Telephone Cable, Rear & Side Mount DIN Rail Adapter Included
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Included Accessories.

RJ45 to Coax Cable RJ45 to 2-Wire Adapter	RJ45 to Coax Adapter Cable For Use With VDSL2 Port & Coaxial Cable RJ45 to 2-Pin Screw Terminal Adapter For Use With VDSL2 Port & 2-Wire Cable (Telephone, Bell, J-Y(ST)Y etc.)
DIN Rail Clip Rear/Side DIN Adapter	Metal DIN Rail Clip & Screws For DIN Rail Mounting AMG173 Series Products Adapter Plate & Screws For Rear Or Side DIN Rail Mounting AMG173 Series Products



Recommended PSUs.

AMGPSU-W12-P25	Plug Top Mounting Light Industrial Grade PSU, 0 to +70°C, 12VDC, 25W, UK/EU/US Plug Heads Included
AMGPSU-I12-P24A	DIN-Rail Mount Industrial Grade PSU, -25 to +70°C, 12VDC, 24W*
* Also available in kit form including mains line cord, DC cable and DIN rail. Order with -K at the end of the part code (e.g. AMGPSU-I12-P24A-K).	